BRAINSTORM GROUP (BSG)

MTH101
Elementary Mathematics I
E-Test Questions (2024)

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1. *Given* $f(x) = x + 3, g(x) = x^2 + 1$ *and* $h(x) = x$. *Find* foh

A. $x^2 + 1$ B. $x^2 + 4$
C. $x + 3$ D. $x^2 + 2$

2. *Given* $f(x) = x + 3, g(x) = x^2 + 1$ *and* $h(x) = x$. *Find* goh

A. $x^2 + 1$ B. $x^2 + 4$
C. $x + 1$ D. $x^2 + 2$

3. *Given* $f(x) = x + 3, g(x) = x^2 + 1$ *and* $h(x) = x$. *Find* gof

A. $x^2 + 6x - 10$ B. $x^2 - 6x - 10$
C. $x^2 - 6x + 10$ D. $x^2 + 6x + 10$

4. *Given* $f(x) = x + 3, g(x) = x^2 + 1$ *and* $h(x) = x$. *Find* hof

A. $x - 3$ B. $x^2 + 1$
C. $x + 3$ D. $x^2 - 1$

5. *Given* $f(x) = x + 3, g(x) = x^2 + 1$ *and* $h(x) = x$. *Find* hog

A. $x - 3$ B. $x^2 + 1$
C. $x + 3$ D. $x^2 - 1$

6. *Given* $f(x) = x + 3, g(x) = x^2 + 1$ *and* $h(x) = x$. *Find* fog

A. $2x^3 + 4$ B. $2x^2 + 4$
C. $x^2 + 4$ D. $2x^2$

7. *Given* $f(x) = x + 3, g(x) = x^2 + 1$ *and* $h(x) = x$. *Find* $fogoh$

A. $2x^3 + 4$ B. $2x^2 + 4$
C. $x^2 + 4$ D. $2x^2$

8. *Evaluate* $|x - 3| > 1$

A. $x < 2$ B. $x > 4$
C. $x > 2$ D. *A & B*

9. *Evaluate* $|2x - 5| < 3$

A. $x > 4$ B. $1 < x < 4$
C. $x > 1$ D. *A & B*

10. *Evaluate* $|2 - 4x| \geq 6$

A. $x \geq 2$ B. $x \leq -1$
C. $2 \leq x > -1$ D. *A & B*

11. *Express* $cos^2\theta$ *in terms of multiole angles*

A. $2[cos2\theta - 1]$ B. $2^{-1}[cos2\theta - 1]$
C. $2[cos2\theta + 1]$ D. $2^{-1}[cos2\theta + 1]$

12. *Find* $f(x) = \sqrt{x + 4}$ *when* $f(0)$

A. -2 B. 0
C. 2 D. $\sqrt{2}$

13. *Find* $f(x) = \sqrt{x + 4}$ *when* $f(-1)$

A. 0 B. $i + 1$
C. $\sqrt{3}$ D. $\sqrt{4}$

14. *Find* $f(x) = \sqrt{x + 4}$ *when* $f(-2)$

A. $\sqrt{3}$ B. 0
C. i D. $\sqrt{2}$

15. *Find* $f(x) = \sqrt{x + 4}$ *when* $f(-3)$

A. i B. $-i$
C. 1^2 D. 0

16. *Find* $f(x) = \sqrt{x + 4}$ *when* $f(-4)$

A. $\sqrt{2}$ B. 0
C. -2 D. i

17. *Given that* $f(x) = \sqrt{x + 4}$. *Compute* $f(x) = 2$

A. 1 B. 0
C. $\frac{1}{\sqrt{2}}$ D. i

18. *Given that* $f(x) = \sqrt{x + 4}$. *Compute* $f(x) = 1$

A. $i + 2$ B. 0
C. 4 D. -3

19. *What is the value of* a *given* $a + ib = \frac{1-i}{2+i}$

A. $-\frac{1}{5}$ B. $\frac{3}{5}$
C. $-\frac{3}{5}$ D. $\frac{1}{5}$

20. Given that $Z_1 = 3 + 2i$ and $Z_2 = 4 - 5i$, evaluate $Z_1 Z_2$

A. $2 - 7i$ B. $22 - 7i$
C. $2 + 7i$ D. $22 + 7i$

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21. *If* $Z_1 = 1 - 3i$ *and* $Z_2 = -2 + 5i$, *determine* $\frac{Z_1 Z_2}{Z_1 + Z_2}$ *in the form* $a + ib$

A. $-\frac{9}{5} + \frac{37}{5}i$ B. $\frac{9}{5} - \frac{37}{5}i$

C. $-\frac{12}{25} - \frac{13}{25}i$ D. $\frac{9}{25} - \frac{37}{25}i$

22. Given the universal set $M = \{1, 2, 3, 4, 5, 6, 7, 8\}$ where $A = \{1, 3, 5, 7\}, B = \{1, 2, 5, 7\}$ *and* $C = \{3, 6, 7, 8\}$ list the members of $A^c \cap B \cap C^c$

A. $\{1\}$ B. $\{2\}$

C. $\{3\}$ D. $\{4\}$

23. If $X \cap Y = \emptyset$ then X and Y are said to be

A. finite sets B. disjoint sets

C. null sets D. infinite sets

24. *Solve* $x^2 + 8x + 16 > 0$

A. $x > -4$ B. $x < 4$

C. $x \geq 2$ D. $x \geq -4$

25. If set $F = \{x : (2 < x \leq 5) \cup (9 < x < 12)\}$. List the members of set F.

A. $\{3, 4, 5, 10, 11\}$ B. $\{3, 4, 5, 7, 11\}$

C. $\{2, 3, 4, 5, 10\}$ D. $\{3, 4, 5, 6, 7\}$

26. Determine the modulus, argument of $2 + 3i$

A. $\sqrt{13},\ 33.69°$ B. $\sqrt{13}, 56.31°$

C. $\sqrt{13}, -33.69°$ D. $\sqrt{13}, -56.31°$

27. Determine the modulus, argument of $2 + 4i$.

A. $2\sqrt{5},\ 63.43°$ B. $2\sqrt{10},\ 63.43°$

C. $2\sqrt{5},\ 26.57°$ D. $2\sqrt{10},\ 26.57°$

28. Solve the inequality $2x - 3 < x + 7$

A. $x + 10 < 0$ B. $x < 10$

C. $x > -10$ D. $x - 10 > 0$

29. Determine the value of x satisfying $|x + 3| \geq 7$

A. $x \leq -4, x \geq 10$

B. $x \leq -10, x \geq 4$

C. $x \leq -10, x \geq -4$

D. $x \leq -4, x \geq 10$

30. *Simplify* $\sqrt{20} + \sqrt{125} + \sqrt{45}$

A. $5(\sqrt{5})$ B. $7(\sqrt{5})$

C. $10(\sqrt{5})$ D. $11(\sqrt{5})$

31. *Simpilfy* $3\sqrt{2} - \sqrt{32} + \sqrt{50}$

A. $5\sqrt{2}$ B. $6\sqrt{2}$

C. $10\sqrt{2}$ D. $4\sqrt{2}$

32. *Find* $\frac{Z_1}{Z_2}$ *if* $Z_1 = 2 - 4i$ *and* $Z_2 = 6 + 7i$

A. $\left(\frac{16}{85} + i\frac{38}{85}\right)$ B. $\left(\frac{15}{85} - i\frac{38}{85}\right)$

C. $-\left(\frac{16}{85} - i\frac{38}{85}\right)$ D. $-\left(\frac{16}{85} + i\frac{38}{85}\right)$

33. *Solve* $4x + \frac{1}{3} \geq \frac{2}{3}x + 2$

A. $x \geq 1$ B. $2x \geq 1$

C. $2x \leq 1$ D. $x \leq 1$

34. Given that $Z_1 = 2 + 3j$ and $Z_2 = 3 - 4j$, evaluate $Z_2 - Z_1$

A. $1 + 7j$ B. $1 - 7j$

C. $-1 + 7j$ D. $-1 - 7j$

35. Given $Z_1 = 1 + 2i,\ Z_2 = 4 - 3i,\ Z_3 = -2 + 3i$, evaluate $|Z_1 + Z_2| - |Z_3| \times \sqrt{2}$

A. $\sqrt{10}$ B. 10

C. $2\sqrt{5}$ D. 0

36. If $Z_1 = 1 + 2i$ *and* $Z_2 = 4 - 3i$, what is the value of $|Z_2 - Z_1|$?

A. $\sqrt{34}$ B. $\sqrt{16}$

C. $2\sqrt{5}$ D. 4

37. Determine the modulus, argument of $-5 - 2i$.

A. $5.385, -158.2°$ B. $5.385,\ 26.57°$

C. $4.583, -158.2°$ D. $4.583,\ 26.57°$

38. *Given* $w = u + iv, z = x + iy$ *and* $w = \frac{1}{z}$. *find* u.

A. $\frac{x}{x^2+y^2}$ B. $-\frac{x}{x^2+y^2}$

C. $\frac{y}{x^2+y^2}$ D. $-\frac{y}{x^2+y^2}$

39. *Given* $w = u + iv, z = x + iy$ *and* $w = \frac{1}{z}$, *find* v.

A. $\frac{x}{x^2+y^2}$ B. $-\frac{x}{x^2+y^2}$

C. $\frac{y}{x^2+y^2}$ D. $-\frac{y}{x^2+y^2}$

40. _____ numbers ranges from $-\infty$ to $+\infty$

A. real B. integer

C. complex D. natural

41. In a village all the people speak Hausa or English or both. If 97% speak Hausa and 64% speak English, what percentage speak both languages?

A. 61% B. 63%

C. 67% D. 69%

42. In a village of 650 youths, 250 are girls, 425 of the youths like music. If 90 girls do not like music, how many boys do not like music?

A. 135 B. 160 C. 175

D. 310 E. 335

43. In a class of 20 boys, 16 play soccer, 12 play hockey and 2 are not allowed to play games. Find how many students play soccer and hockey.

A. 6 B. 8

C. 10 D. 12

44. Given that $P = \{1,\ 3,\ 5,\ 7\}$ *and* $Q = \{2,\ 5,\ 6,\ 8,\ 9\}$. *Find* $(P \cup Q)$.

A. $\{1, 2, 3, 5, 6, 7, 8, 9\}$

B. $\{1, 2, 3, 6, 7, 8, 9\}$

C. $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

D. $\{3, 5\}$

45. The sets $P = \{2, 5, 8\}$ and $Q = \{2, 3, 5, 7\}$ are subsets of the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Find $P' \cap Q$

A. $\{3\}$ B. $\{3, 5\}$

C. $\{3, 7\}$ D. $\{1, 3\}$

46. The sets $P = \{2, 5, 8\}$ and $Q = \{2, 3, 5, 7\}$ are subsets of the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.
Find $n(P \cap Q')$

A. 1 B. 2

C. 3 D. 0

47. Given $f(x) = 2x^2$ and $g(x) = 3x + 1$, Compute $(fog)(2)$

A. 25 B. 49

C. 98 D. 89

48. If $f(x) = 2(x-3)^2 + 3(x-3) + 4$ and $g(y) = 5 + y$. Find $f[g(-1)]$

A. 2 B. 3

C. 4 D. 9

49. The function of f and g are defined over R as $f: x = x^2 + 2$ and $g: x = \frac{2x}{1-x}$. Find $fog(3)$.

A. -3 B. -9

C. 9 D. 11

50. If $f: x = 2x - 3$ and $g: x = 3x + p$, for what value of constant p is $fog = gof$.

A. -3 B. 3

C. -2 D. -6

51. Given $f(x) = 2x^2$ and $g(x) = 3x + 1$, Compute (fog).

A. $9x^2 + 6x + 1$

B. $18x^2 + 12x + 1$

C. $18x^2 + 12x + 2$

D. $18x^2 + 12x - 2$

52. *Given* $Z = 4 - 2\sqrt{2}i$, *find* $Z\bar{Z}$

A. 21 B. 26

C. 24 D. 28

53. Determine the modulus, argument of $i(2 - i)$.

A. 1.732, 63.43°

B. 2.236, 63.43°

C. 1.732, 26.57°

D. 2.236, 26.57°

54. Find the range of values of x for which the inequality $\frac{1}{3}x - 2 \leq \frac{2}{5}x + \frac{1}{4}$

A. $x \geq -\frac{4}{135}$ B. $x \geq -\frac{135}{4}$

C. $x \geq \frac{4}{135}$ D. $x \geq \frac{135}{4}$

55. *Given the function* $f(x) = 5x - 2$, *find* $f^{-1}(-1)$

A. $\frac{1}{5}$ B. -7

C. $-\frac{1}{7}$ D. $-\frac{1}{5}$

56. *Find the inverse of* $f: x = \frac{x+1}{x-2}$

A. $\frac{2x+1}{x-1}$ B. $\frac{2x-1}{x-1}$

C. $\frac{2x+1}{x+1}$ D. $\frac{x-1}{2x+1}$

57. *Find the solution of the inequality* $(12 - 4x) \geq 6$

A. $x \geq \frac{3}{2}$ B. $x \geq -\frac{3}{2}$

C. $x \leq \frac{3}{2}$ D. $x \leq -\frac{3}{2}$

58. *If* $X = \{2, 4, 6\}$, *determine the number of power set* X *denoted as* $nP(X)$.

A. 3 B. 6

C. 8 D. 16

59. *Suppose set* $R = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$ *find the number of element of* R.

A. 2 B. 8

C. 10 D. 20

60. *If* $z = cos\theta + isin\theta$, *then find the value of* $\left(z + \frac{1}{z}\right)^2$ *is*

A. $2cos\theta$ B. $2isin\theta$

C. $4cos^2\theta$ D. $-4sin^2\theta$

A. *The discriminant for the equation* $3x^2 - 4x + 12 = 0$ *is* ___

A. 72 B. -72

C. 128 D. -128

Solution

Comparing the equation with the general equation: $ax^2 + bx + c = 0$, *we have*

$a = 3, b = -4$ & $c = 12$

Discriminant, $\Delta = b^2 - 4ac$

$\Delta = (-4)^2 - 4(3)(12) = 16 - 144$

$\Delta = \mathbf{-128}$ [***ANSWER is D***]

B. *Determine the discriminant of the quadratic equation* $x^2 + 4x = 4$

A. 0 B. 16

C. 32 D. -16

Solution

$x^2 + 4x = 4 \implies x^2 + 4x - 4 = 0$

Comparing the equation with the general equation: $ax^2 + bx + c = 0$, *we have*

$a = 1, b = 4$ & $c = -4$

Discriminant, $\Delta = b^2 - 4ac$

$\Delta = (4)^2 - 4(1)(-4) = 16 + 16$

$\Delta = \mathbf{32}$ [***ANSWER is C***]

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ANSWERS

1.	C	11.	D	21.	B	31.	D	41.	A	51.	C
2.	A	12.	C	22.	B	32.	D	42.	A	52.	C
3.	D	13.	C	23.	B	33.	B	43.	C	53.	B
4.	C	14.	D	24.	A	34.	B	44.	A	54.	B
5.	B	15.	C	25.	A	35.	D	45.	C	55.	A
6.	C	16.	B	26.	B	36.	A	46.	A	56.	A
7.	C	17.	B	27.	A	37.	A	47.	C	57.	C
8.	D	18.	D	28.	B	38.	A	48.	D	58.	C
9.	B	19.	D	29.	B	39.	D	49.	D	59.	C
10.	D	20.	B	30.	C	40.	A	50.	D	60.	C

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